

Assignment 1

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Course: Theory of Automata

Task: Assignment 1

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Q1. let $S(n) = 1 + 2 + \dots + n$ be the sum of first n natural numbers and let $C(n) = 1^3 + 2^3 + \dots + n^3$ be the sum of first n cubes. Prove that

$C(n) = S(n)^2$. Prove the following equalities by induction on n , to arrive at the conclusion that $C(n) = S(n)^2$.

(a) (05 points) $S(n) = \frac{n(n+1)}{2}$

using $n=1$

$$\text{L.H.S} = S(n) = 1 + 2 + \dots + n \\ = 1$$

$$\text{R.H.S} = S(n) = \frac{n(n+1)}{2} = \frac{1(1+1)}{2} \\ = \frac{1(2)}{2} = \frac{2}{2} = 1$$

$$L.H.S = R.H.S \quad \checkmark$$

using $n=k$

$$\cancel{L.H.S} \Rightarrow \cancel{S(k)} = \cancel{1+2+\dots+k}$$

I assume equality holds for ~~k~~

$$S(k) = \frac{k(k+1)}{2}$$

$n=1$ is the base case
where equality holds

using $n=k+1$

now for any arbitrary k ,

$$S(k+1) = S(k) + (k+1)$$

Sum till $(k+1)$ = Sum till k + the $(k+1)$ th value

$$= \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{k^2 + k}{2} + (k+1)$$

$$= \frac{k^2 + k + 2(k+1)}{2}$$

$$= \frac{k^2 + k + 2k + 2}{2}$$

$$= \cancel{k^2 + 3k + 2}$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

This is equal to

$$S(n) = \frac{n(n+1)}{2}$$

$$S(k+1) = \frac{(k+1)(k+1+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

hence LHS = RHS

equality holds for $k+1$

hence by induction,

$$S(n) = \frac{n(n+1)}{2}$$

(b) (05 points) $c(n) = \frac{n^2(n+1)^2}{4}$

using $n=1$

$n=1$ is the base case

$$\begin{aligned} \text{LHS} = c(n) &= 1^3 + 2^3 + \dots + n^3 \\ &= 1^3 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{RHS} = c(n) &= \frac{n^2(n+1)^2}{4} \\ &= \frac{1^2(1+1)^2}{4} \\ &= \frac{1(2)^2}{4} = \frac{4}{4} = 1 \end{aligned}$$

$$\text{L.H.S} = \text{R.H.S} \quad \checkmark$$

$$1 = 1 \quad \checkmark$$

Using $n=k$

I assume k holds equality.

$$c(n) = \frac{n^2(n+1)^2}{4}$$

$$C_8(k) = \frac{k^2(k+1)^2}{4}$$

using $n=k+1$

$$C_8(k) = \frac{k^2(k+1)^2}{4}$$

$$C_8(k+1) = \frac{(k+1)^2 \cancel{(k+1)^2} (k+1+1)^2}{4}$$

$$= \frac{(k+1)^2 (k+2)^2}{4}$$

$$C_8(k+1) = C_8(k) + (k+1)$$

$$= \frac{k^2(k+1)^2}{4} + k+1$$

$$= \frac{k^2(k+1)^2 + 4(k+1)}{4}$$

$$= \frac{\cancel{k^2(k+1)^2} + 4(k+1)}{4}$$

$$= \frac{(k+1)[k^2(k+1) + 4]}{4}$$

$$= \frac{(k+1)[k^3 + k^2 + 4]}{4}$$

$$\frac{k^2(k+1)^2 + 4(k+1)}{4}$$

$$\frac{k^2(k^2 + 2k + 1) + 4k + 4}{4}$$

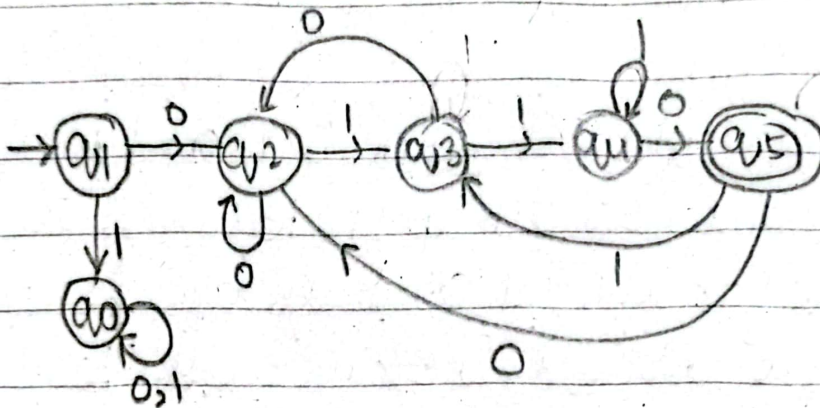
$$\frac{k^4 + 2k^3 + k^2 + 4k + 4}{4}$$

take $y = k^2$
 $y^2 + 2y + 4$

Q2. For each of the following, draw a state diagram of a DFA that recognizes the specified language. In all cases $\Sigma = \{0, 1\}$

(a) (105 points)

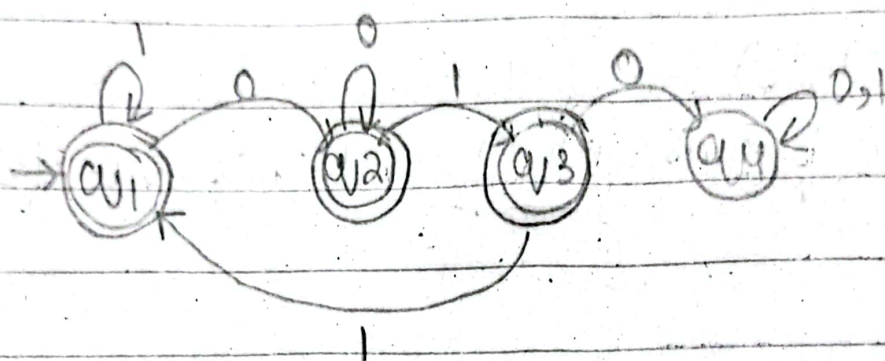
$L = \{w : w \text{ begins with } 0 \text{ and ends with } 110\}$



- So ~~if~~ I start at q_1 . If I get a 0, I move forward, otherwise I go to a trap state which is not an accepting state, as the requirement is to start with 0. Then, after 0, if I get a 110, I go from q_2 to q_3 to q_4 to q_5 which is an accepting state. If after 110 I get more digits, I will recheck for 110. If in between checking 110, I get a 0 after the first 1, I will recheck for 1. If I get a ~~0~~ 1 after the second 1, I will stay there and wait for a 0 to come.

(b) (05 points)

$L = \{ w : w \text{ does not contain } 010 \text{ as substring} \}$



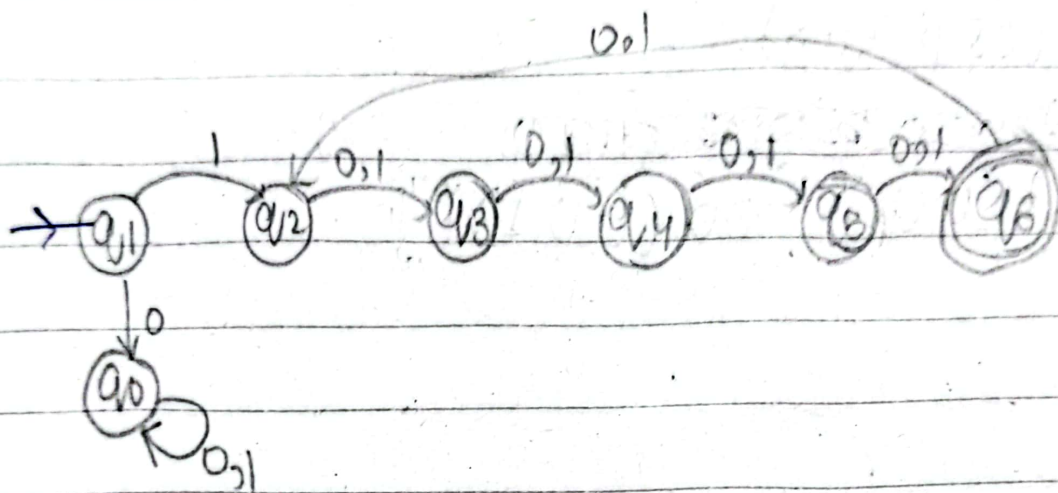
So I start at q_1 . If I get a 0 then I then 0, I end up at a trap state which is nonaccepting state q_4 and all alphabets $\{0,1\}$ I get at q_4 , I stay at q_4 . The rest of the states accept as I ~~got~~ didn't get a 010 substring yet.

If after a 01 I get another 1, I have to recheck for 0 so I move back a state.

— All the rest are accepting states
EXCEPT the state which has read 010.

(c) (05 points)

$L = \{ w : w \text{ starts with 1 and has length multiple of 5} \}$



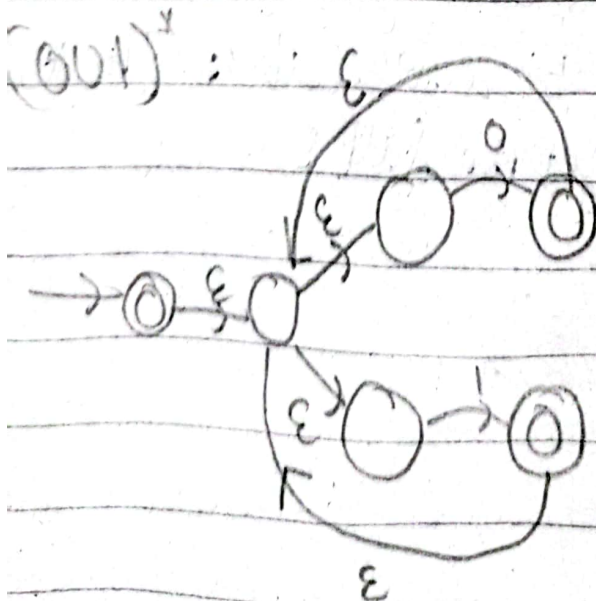
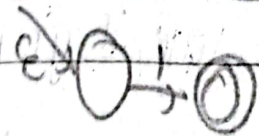
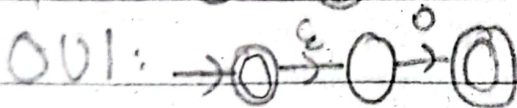
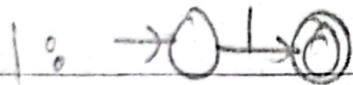
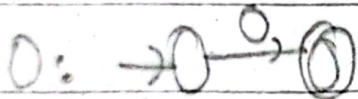
- so if I get a 1 at my start state, I go to a trap state. If I get any 5 letters, I go to accepting state q6. If there are letters after it, I recheck if again I get the full 5.

Q3. For each of the following, draw a state diagram of a DFA that recognizes the specified language. In all cases $\Sigma = \{0, 1\}$.

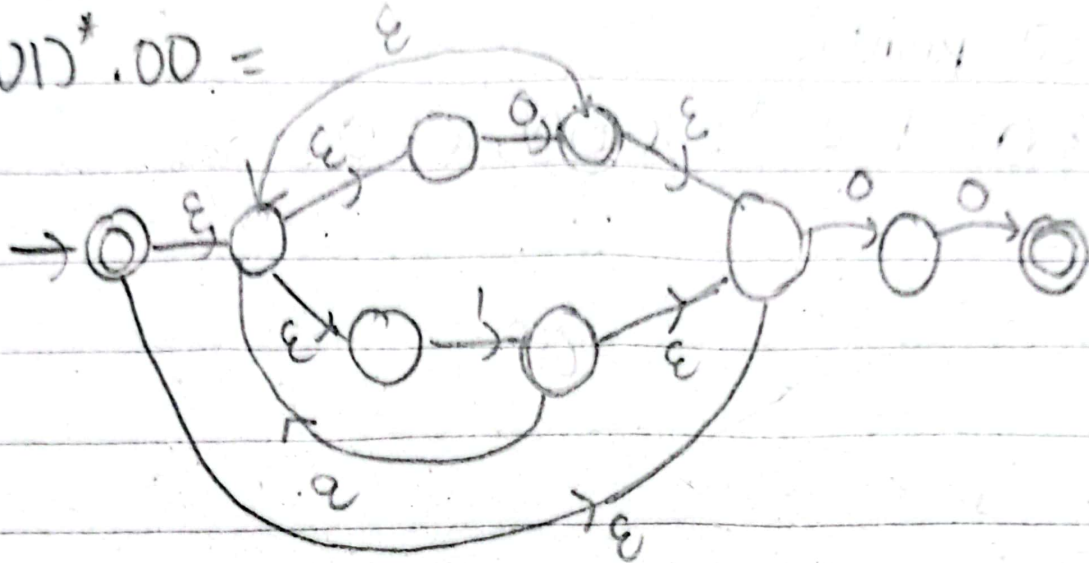
(a) (05 points)

~~Strings begin with 0~~

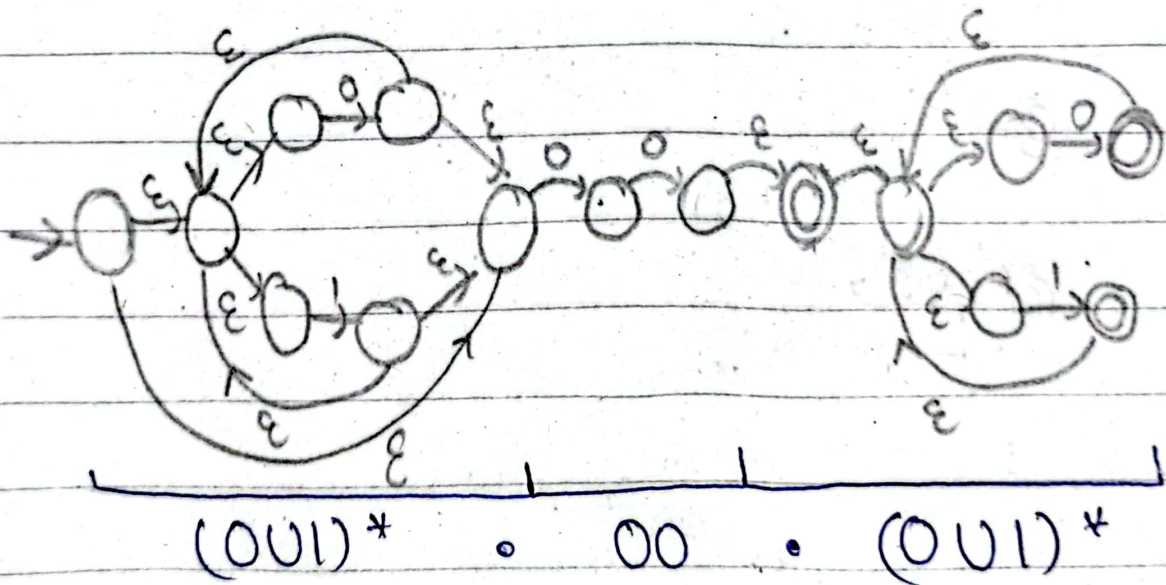
$$L = \{(0 \cup 1)^* 00 (0 \cup 1)^*\}$$



$$(0 \cup 1)^* \cdot 00 =$$



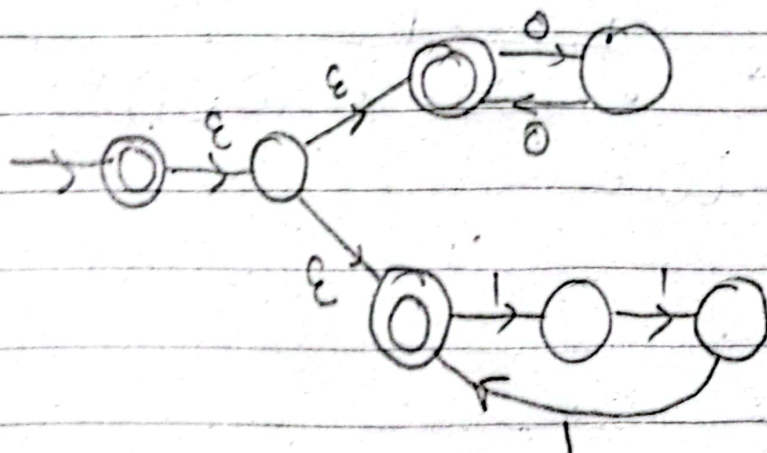
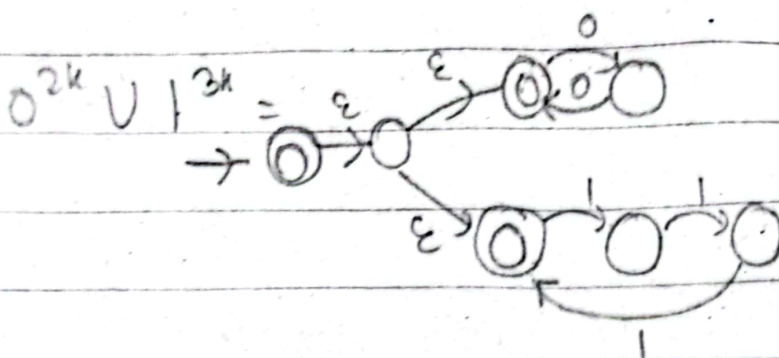
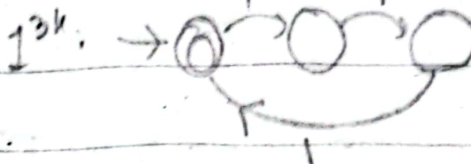
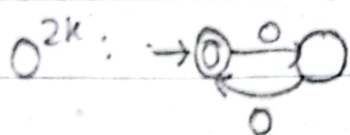
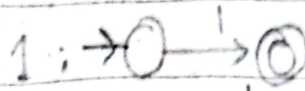
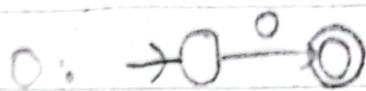
$$(0 \cup 1)^* 00 (0 \cup 1)^*$$



- It accepts $\epsilon \cdot 00 \cdot \epsilon$

(b) (05 points) ^{3.2.6}

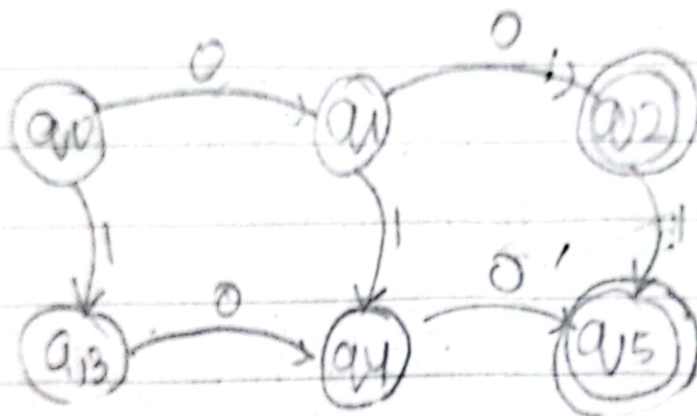
$$L = \{0^{2k} \cup 1^{3k} : k \in \mathbb{N}, k \geq 0\}$$



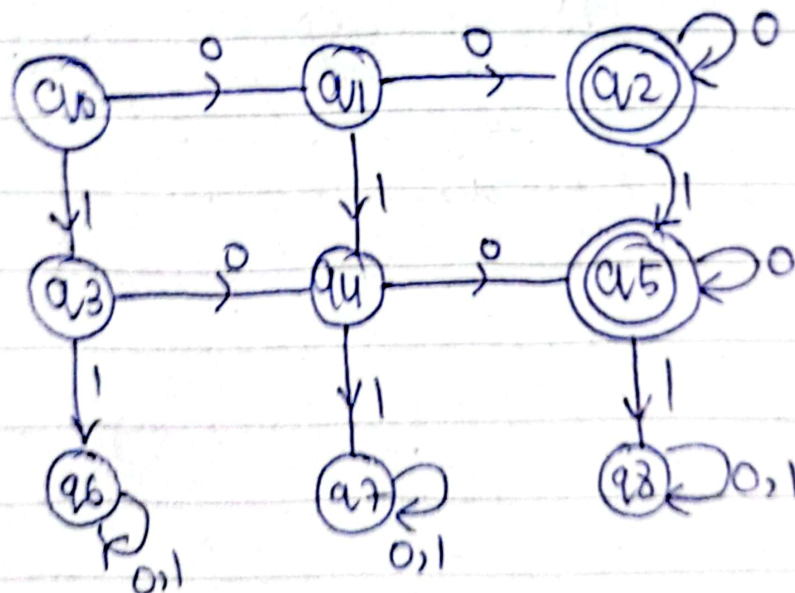
This can also accept ϵ as if $k=0$, nothing would exist $[\epsilon]$ and it would be accepted.

(c) (05 points)

$L = \{ w : w \text{ contains at least two 0's and at most one 1} \}$

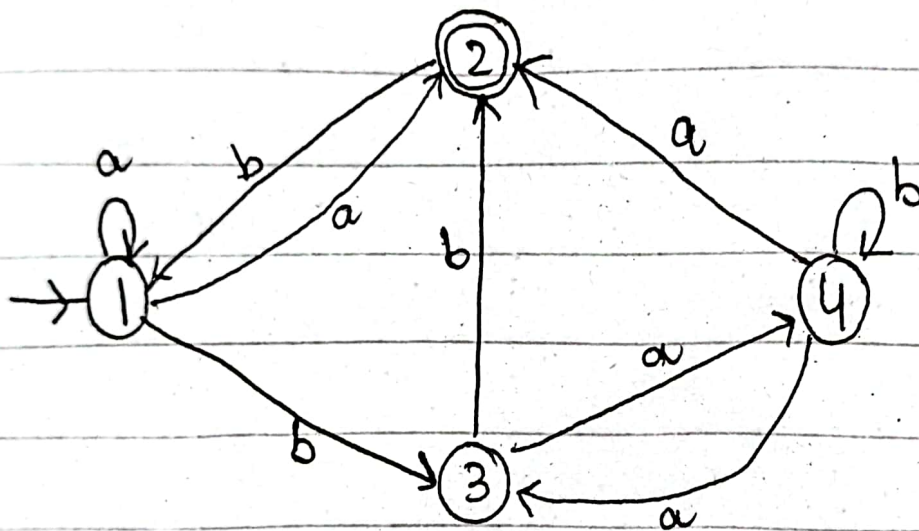


This is an NFA.

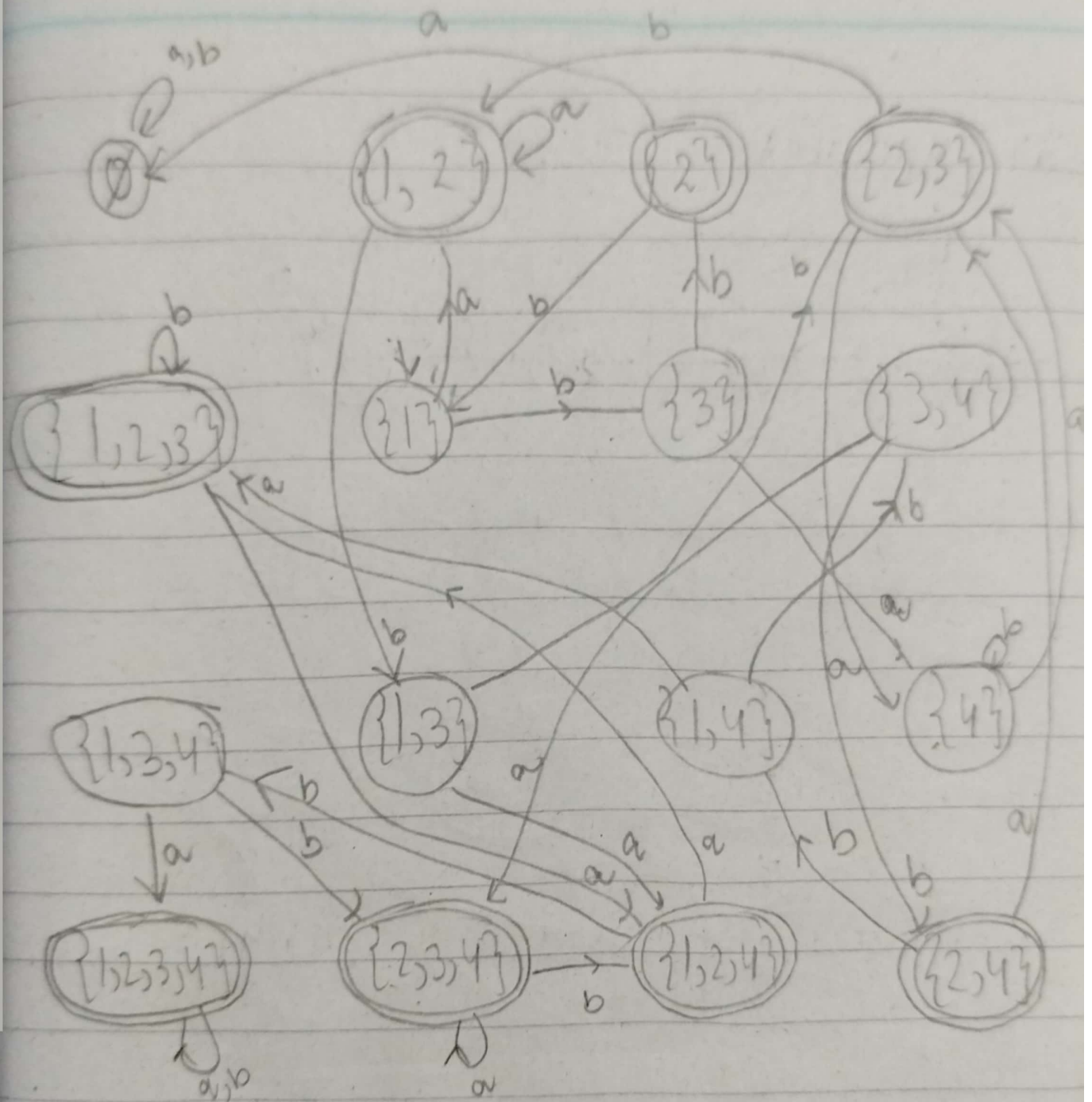


This is the DFA of the same L .

Q4. Convert the following NFA's to equivalent DFA's

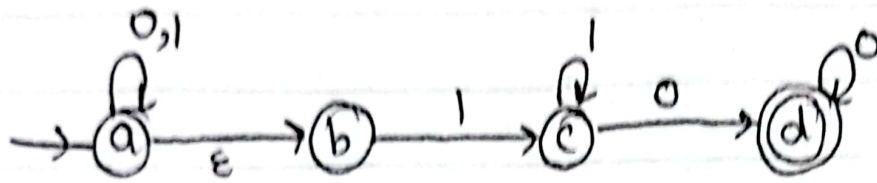


~~$\emptyset, \{1, 2, 3, 4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\},$~~
 ~~$\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \{1, 2, 3, 4\}$~~

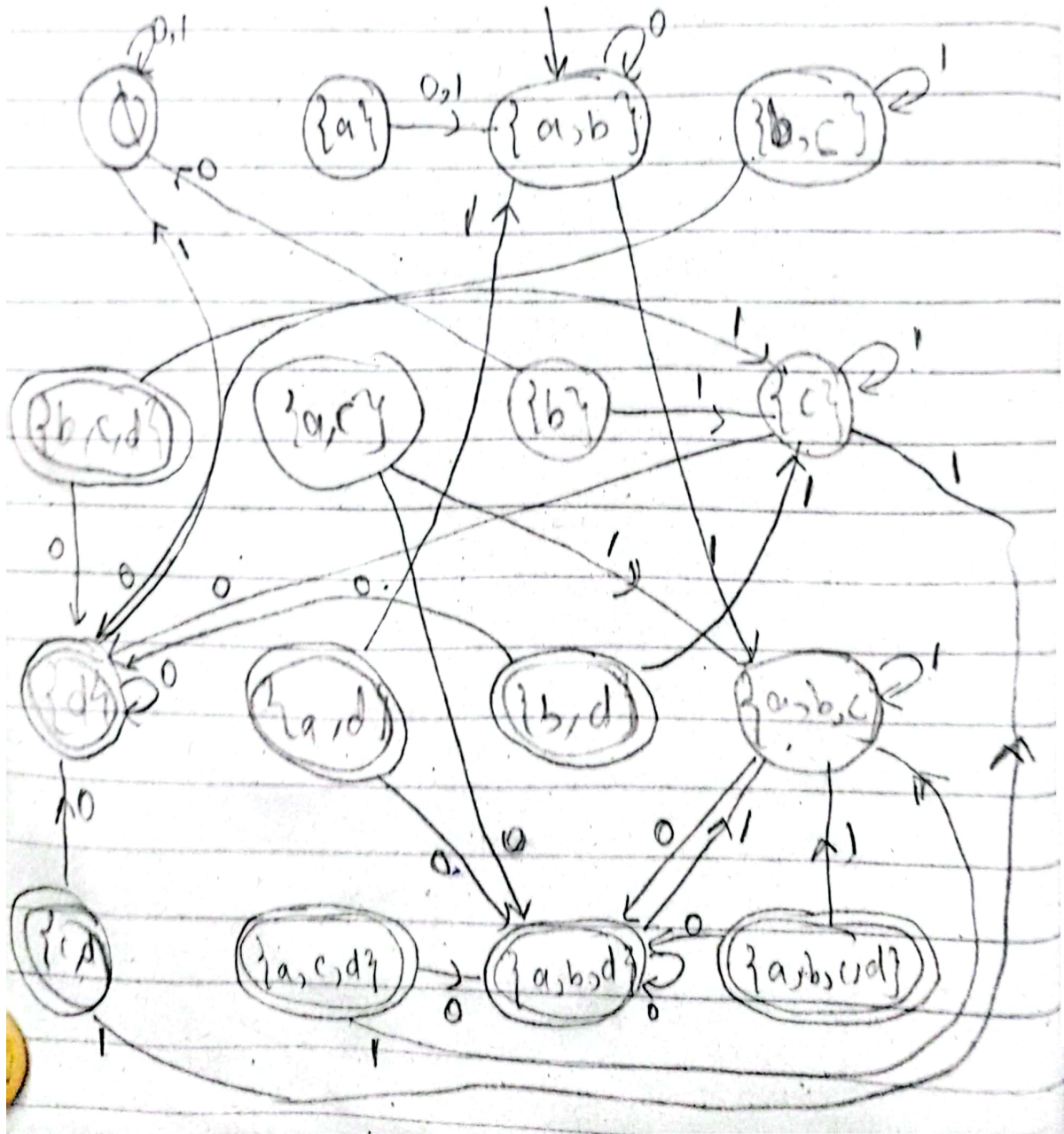


(a) (05 points)

$\{0,1\}$

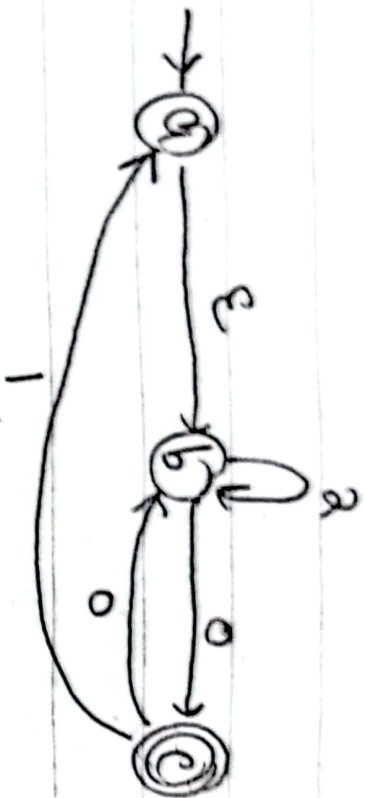


~~$\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a,b\}, \{a,c\}, \{a,d\}, \{b,c\}, \{b,d\}, \{c,d\},$~~
 ~~$\{a,b,c\}, \{a,c,d\}, \{a,b,d\}, \{b,c,d\}, \{a,b,c,d\}$~~

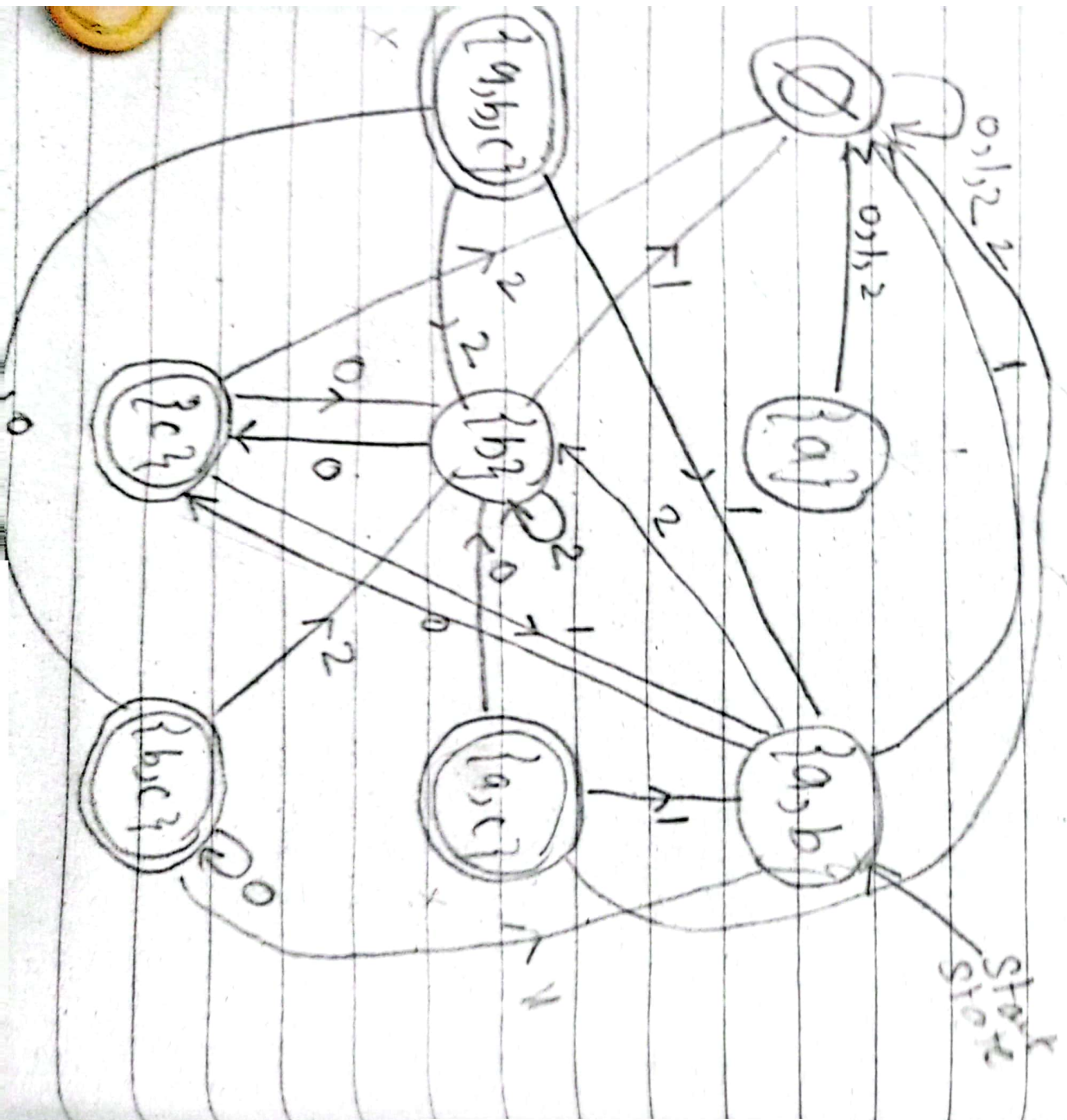


(b) (05 points)

10, 1, 2, 3



$\emptyset, a, b, c, \{a, b\}, \{a, c\}, \{a, b, c\}, \{b, c\}$



Q5. (05 points)

Show that a finite language is regular,

A language is a set of strings
of $\{s_1, s_2, \dots, s_n\}$
where s_i denotes string and
 $i \geq 0$ to n where n is
a natural number

A language is regular when either

- It can be represented as a regular expression
- or DFA exists for it

- so I have a finite language

$\{s_1, s_2, \dots, s_n\}$

When dividing into sets I can
express it as

$\{s_1\}, \{s_2\}, \dots, \{s_n\}$

or
 $\{s_1\} \cup \{s_2\} \cup \dots \cup \{s_n\}$

hence as according to Theorem 1,

union is endorsed for finite languages
hence it is a regular language.

Q6. (05 points)

Let L be a regular language. Show that the following language L' is also regular,

$$L' = \{ x : \text{there exists } y \text{ such that } xy \in L \}$$

Let

$L(M)$ where

$M = (Q, \Sigma, \delta, s, F)$ be a DFA accepting L .
WLOG, we assume all states of M are reachable from start state.

To show that $\text{Prefix}(L)$ is regular, we find a DFA $\bar{M}$ that will accept it.

We find that $\bar{M} = (Q, \Sigma, \delta, s, \bar{F})$ that is the DFA that accepts the set of prefixes is the same as the one for L except we only change the set of accepting states.

We say that $q \in \bar{F}$ if and only if there is a path from q to an accepting state f of the original machine M .

In other words, there is a string $y \in \Sigma^*$ so that $\delta(q, y) = f \in F$.

This works because: if $\delta(s, x) = q$, then $x \in \text{Prefix}(L)$ since $\delta(s, xy) = f \in L$ and conversely.

Informally, the set of accepting states for $\text{Prefix}(L)$ consist of any state which can be joined to an accepting state of the original machine.

We now consider a generalization of the set of prefixes.

Let L_1 and L_2 be languages over Σ . Then the right quotient of L_1 with L_2 is defined as

$$L_1/L_2 = \{x : xy \in L_1, \text{ for some } y \in L_2\}$$

- If L is regular, then it is accepted by a DFA $M(Q, \Sigma, \delta, s, F)$. Also u is a prefix of $uv \in L$ if and only if there is a path in M from s to $\delta(s, u)$ and there is a path in M from $\delta(s, u)$ to a state in F . Let $S \subseteq Q$ be the states such that for each $q \in S$, there is a path in M from s to q and there is a path in M from q to a state of F . Language $\{u \mid \exists v (uv \in L)\}$ is accepted by DFA $(Q, \Sigma, \delta, s, F \cup S)$ and hence it is regular.

Q7. (10 points)

We say a state $q \in Q$ is a deadend if $S^*(q, w) \cap F = \emptyset$ for all $w \in \Sigma^*$. Argue that we can without loss of generality assume that if q is a deadend, then it is also a trap state. A trap state is one from which you can never escape. Argue that we can assume that a DFA has at ~~least~~ most one trap state. Finally, show that a regular language L is accepted by a DFA with no trap state if and only if $\text{PREFIX}(L) = \Sigma^*$.

[Note: $\text{PREFIX}(L)$ is the set of all prefixes of strings in L . For example, if $L = \{ab, abc, b\}$, then $\text{PREFIX}(L) = \{\epsilon, a, ab, abc, b\}$.]

So if q is a deadend, it is a state which is not an accepting state. A deadend is a state that loops all the alphabets to itself meaning we cannot get out of it. Hence we are trapped; therefore it is proved that a deadend is a trap state.

- So if a DFA has one trap state, and we introduce a second trap state, it is additional and non-necessary. We can connect the second trap state transitions to the first trap state thus eliminating the second trap state.

- So Σ^* means all possible combinations of L . It is an infinite language as because of star (*)

- $\text{PREFIX}(L)$ means all the strings in L , AND all the possible prefixes of all the strings in L .

- If $\text{PREFIX}(L) = \Sigma^*$

That means my $\text{PREFIX}(L)$ contains all possible combinations of my characters/letters in L . That means if I have a machine M which is a ~~PFA~~ finite automaton, all the states will be accepting states as it would accept all possible combinations of more letters [any string would be accepted].

- so now we have established the fact that when we have Σ^* , all states are accepting states.

- ~~the~~

- when all states are accepting states, there will be no trap state.

↓
a trap state is a non accepting state we cannot go out of.

- Therefore when

$$\text{Prefix}(L) = \Sigma^*$$

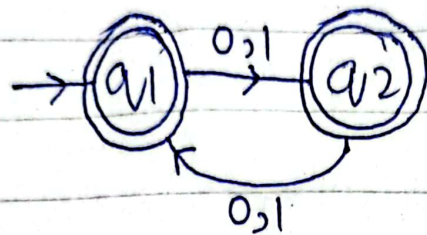
- all strings are accepted
- all states are accepting
- there are zero trap states.

- THEREFORE we have proved that when $\text{Prefix}(L) = \Sigma^*$, there exists a DFA for language L with no trap state.

This was a general proof.
I will proof this more with examples.

example, $\Sigma = \{0, 1\}$

$$\Sigma^* = \{\epsilon, 0, 1, 00, 11, 01, 10, 0001, \dots\}$$



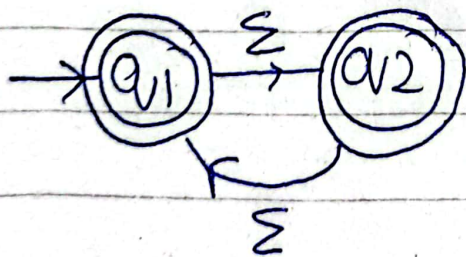
no trap state ✓

$$\text{Prefix}(L) = \{\epsilon, 0, 1, 00, 11, 01, 10, 0001, \dots\}$$

example, $\Sigma = \{a, b, c, d, \dots, z\}$

[it is a finite sigma]

$$\Sigma^* = \{\epsilon, a, b, c, d, \dots, z, aa, bb, cc, \dots, zz, ab, bc, cd, \dots, yz, za, zz a, bbc, \dots\}$$



where Σ is the entire letters of language L .

$$\text{Prefix}(L) = \Sigma^*$$